LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 1: Linear Search

Module 1 | 45 minutes | Searching algorithms | Grades 6–8 · ages 12–14

Objective: Implement linear search to find an item in a list, report its position or “not found”, and count the steps it takes.

Success criteria

- I can implement linear search.
- I can handle the not-found case.
- I can measure an algorithm’s steps.

Key vocabulary

- **linear search**: Checking items one by one in order.
- **target**: The value being searched for.
- **comparison**: Checking if two values are equal.
- **worst case**: The input that makes an algorithm do the most work.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Hidden cards (5 min): 10 face-down number cards; find the 7 by flipping one at a time and counting flips.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Find a friend: learners follow the steps.
14 min	**Perform**	Code Lab: linear search, Worst case
9 min	**Probe**	Code Lab: count the steps, Search in Python, activity
4 min	**Wrap-up**	Exit ticket: What are the two stopping conditions for linear search?

Unplugged warm-up

Hidden cards (5 min): 10 face-down number cards; find the 7 by flipping one at a time and counting flips.

Teaching the idea

Linear search scans sequentially, terminating on match or end; must handle not-found. Efficiency measured by comparisons: worst case n (O(n)). Python bridge shows return -1 and 0-based indexing.

Analogy: Looking for your friend in a queue by checking each face from the front.

Live demo: Search for the first, last and missing names; tabulate steps.

Questions to ask

Ask	Listen for
When does the loop stop?	Found, or past the end.
What is the worst case?	Last item or missing.
Why handle “not found”?	Algorithms must cope with all inputs.

Differentiation

Support
Provide the loop condition; learners write the comparison.

Stretch
Count how many times a name appears (don’t stop early).

Common misconceptions

Misconception	How to fix it
Search always finds something	Plan for not found.
Checking all items is needed even after finding	Stop early.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
What are the two stopping conditions for linear search?

Homework
Time yourself finding a word in a dictionary by reading every word from the start (just for 1 minute!). What would be faster?

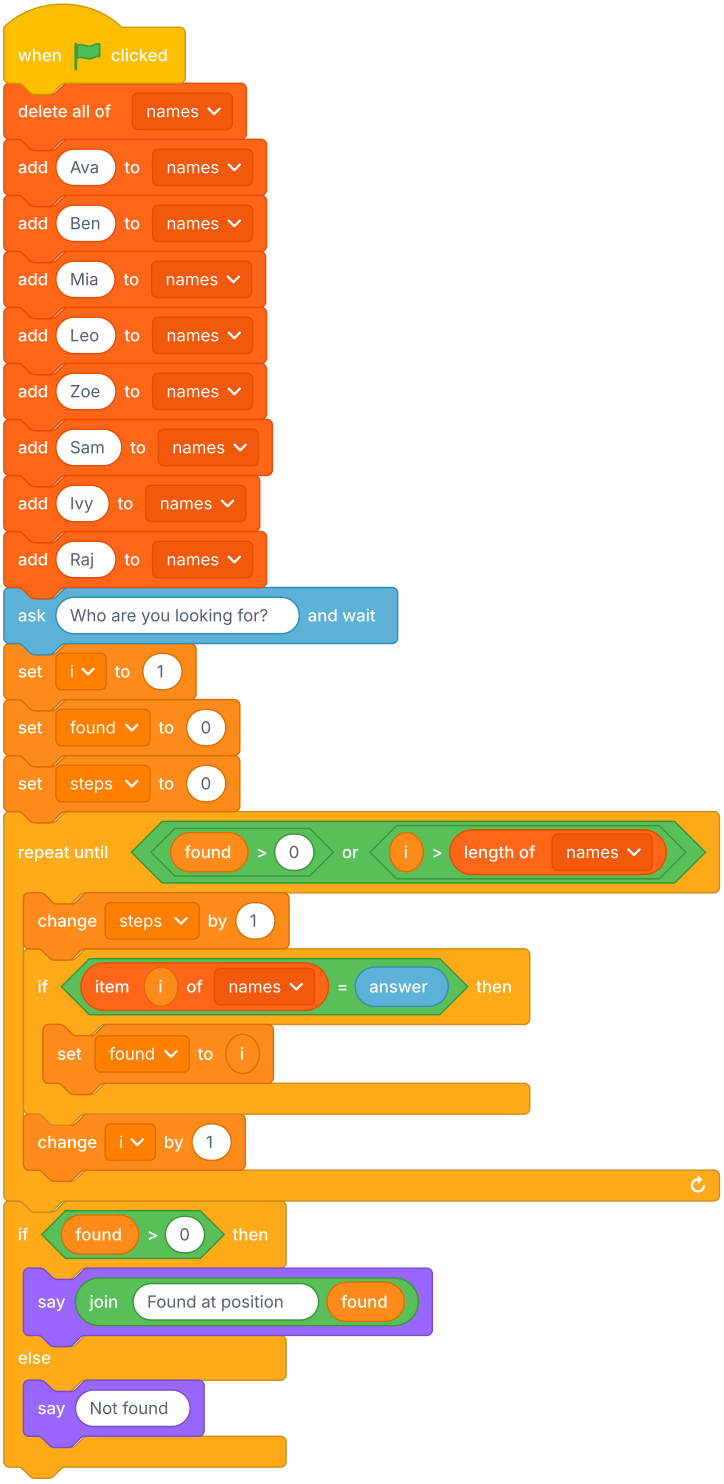
Answer key — Kitabash activities

1. Code Lab: linear search (perform · code lab)

Goals checked: Finds Mia at 3 · Finds the last name · Handles missing names

One solution:

Seeker



2. Worst case (perform · multiple choice)

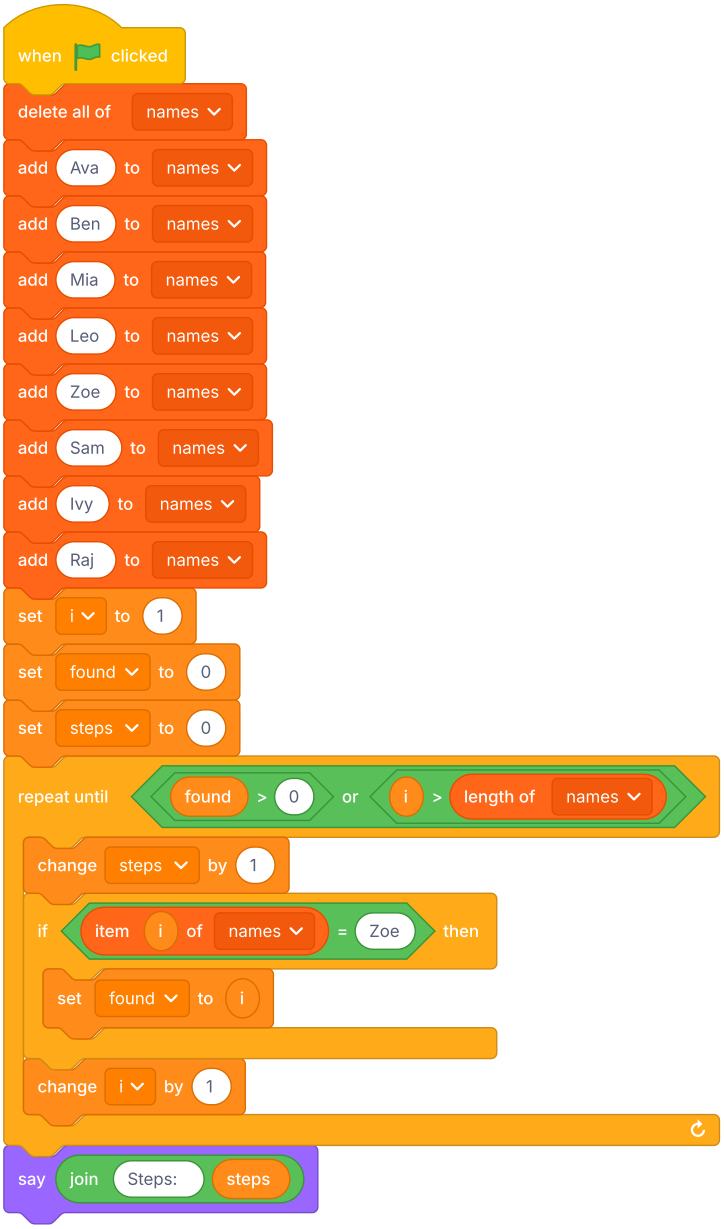
Answer: 1,000

3. Code Lab: count the steps (probe · code lab)

Goals checked: Finds Zoe · Counts 5 steps

One solution:

Seeker



4. Search in Python (probe · code output)

5. Explain your thinking (probe · free response)

Look for: every, items, large. **Example:** Linear search might have to check every item, so with a million items it could take a million comparisons. It is still the right choice when the list is short or not sorted, because faster searches like binary search need sorted data.

Answer key — worksheet

1. **Key words:** linear search = Checking items one by one in order. · target = The value being searched for. · comparison = Checking if two values are equal. · worst case = The input that makes an algorithm do the most work.
2. **Plan your code:** Goals: Finds Mia at 3; Finds the last name; Handles missing names.
3. **Worst case:** A. 1,000 — If the target is last or missing, it checks every item.
4. **Plan your code:** Goals: Finds Zoe; Counts 5 steps.
5. **Search in Python:** A. 2

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.